Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 26)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 1 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

0 1 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 2 . 1 Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

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0 2 . 2 Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

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0 2 . 3 Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

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0 2 . 4 Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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0 3 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 3 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 3 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 4 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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0 4 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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0 4 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

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Blue whale .....

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Large blue butterfly .....

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0 5 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition .....

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Soil maintenance .....

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Interspecies relationships .....

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0 5 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

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0 5 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

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0 6 Table 1 shows features of some fishing methods.

0 6 . 1 Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

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0 6 . 2 Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

0 6 . 3 Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

10

0 7 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

0 7 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

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0 7 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

0 7 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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0 8 A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 8 . 1 Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 8 . 2 Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

0 8 . 3 The test gave p = 0.02. Explain what this means.

[2 marks]

0 8 . 4 Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

0 8 . 5 Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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0 9 . 1 Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 9 . 2 Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

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1 0 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

OR

11 . 2 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS